

Task 3 – Causality Assessment

Pharmacovigilance Internship Submission

1. Case 1: Naranjo Algorithm

Patient Information

Patient: Male, 45 years

Medical History: Hypertension; no known drug allergies

Suspected Drug: Amoxicillin 500 mg orally, three times daily

Indication: Acute bacterial sinusitis

Concomitant Medication: Amlodipine 5 mg once daily, unchanged for 1 year

Adverse Reaction and Timeline

Adverse Reaction: Generalized itchy skin rash

Timeline: Reaction developed approximately 2 days after starting amoxicillin

Action Taken: Amoxicillin was discontinued on June 12, 2026

Dechallenge: Rash and itching improved within 24 hours and resolved completely by June 15, 2026

Rechallenge: Not performed

Alternative Causes: No new medicines, foods, cosmetics, or illnesses were reported

Naranjo Adverse Drug Reaction Probability Scale

#	Question	Answer	Score
1	Are there previous conclusive reports on this reaction?	Yes	+1
2	Did the adverse event appear after the suspected drug was administered?	Yes	+2
3	Did the adverse reaction improve when the drug was discontinued or a specific antagonist was administered?	Yes	+1
4	Did the adverse reaction reappear when the drug was readministered?	Do not know	0
5	Are there alternative causes that could have caused the reaction?	No	+2
6	Did the reaction reappear when a placebo was given?	Do not know	0
7	Was the drug detected in the blood/other fluids in a concentration known to be toxic?	Do not know	0
8	Was the reaction more severe when the dose was increased, or less severe when the dose was decreased?	Do not know	0
9	Did the patient have a similar reaction to the same or similar drugs in any previous exposure?	No	0
10	Was the adverse event confirmed by any objective evidence?	Do not know	0

Total Score Calculation: +1 +2 +1 +0 +2 +0 +0 +0 +0 +0 = 6

Interpretation: 5–8 = Probable

Final Classification: Probable

Brief Rationale

The reaction followed a clear temporal sequence after drug initiation and improved after withdrawal. No alternative cause was identified. Rechallenge was not performed and no objective confirmatory evidence was provided, so the score supports a Probable rather than Definite relationship.

Case 1 Conclusion

Suspected Drug: Amoxicillin

Adverse Reaction: Generalized itchy skin rash

Assessment Method: Naranjo Algorithm

Total Score: 6

Final Classification: Probable

2. Case 2: WHO-UMC Causality Assessment

Patient Information

Patient: Female, 28 years

Diagnosis: Acute bacterial pharyngitis

Suspected Drug: Azithromycin 500 mg orally once daily for 3 days

Concomitant Medication: Oral contraceptive, unchanged for 2 years

Adverse Reaction and Timeline

Adverse Reaction: Nausea and mild abdominal discomfort

Timeline: Symptoms developed on Day 2 of azithromycin treatment

Dechallenge: Symptoms improved after completion/discontinuation of the short course and resolved completely the following day

Rechallenge: Not performed

Alternative Causes: No other new medicines or illnesses were reported

WHO-UMC Causality Assessment

Time Relationship: Reasonable and plausible; the reaction began during treatment on Day 2

Alternative Explanations: None identified from the case information

Response to Withdrawal: Symptoms improved after the treatment course ended and resolved fully the next day

Rechallenge: Not performed

Clinical Evidence: No laboratory or diagnostic confirmation was obtained; the clinical presentation is compatible with the reported reaction

Overall Assessment: Probable / Likely

Rationale

There is a reasonable temporal relationship between azithromycin exposure and the onset of the reaction. Symptoms improved after the treatment course ended, and no alternative explanation was identified. Since rechallenge was not performed, the available evidence supports a Probable/Likely relationship rather than Certain.

Case 2 Conclusion

Suspected Drug: Azithromycin

Adverse Reaction: Nausea and mild abdominal discomfort

Assessment Method: WHO-UMC Causality Assessment

Final Classification: Probable / Likely

3. Overall Conclusion

The two mock cases demonstrate causality assessment using two different approaches. Case 1 applies the Naranjo Algorithm as a numerical scoring method, while Case 2 applies the WHO-UMC criteria-based approach.

In Case 1, the Naranjo score was 6, placing the amoxicillin-associated skin rash in the Probable category. The assessment was supported by the temporal relationship, improvement after withdrawal, and absence of an identified alternative cause.

In Case 2, the WHO-UMC assessment was Probable/Likely. The reaction had a reasonable time relationship with azithromycin, improved after the treatment course ended, and no alternative explanation was identified.

Overall, these cases show how available clinical information can be assessed systematically to determine the strength of the relationship between a suspected drug and an adverse reaction.

Key Learning Point

Causality assessment should be based on the evidence available in the individual case, including the timing of the reaction, response to withdrawal, alternative explanations, and any available supporting evidence.